

Ibna Kowsar

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Research Interests: Mechanistic interpretability (sparse autoencoders, feature attribution), large language models, and representation learning, with applications to vision-language models, structured/clinical (EHR) data, and health informatics.

Education

University of California, Davis

Ph.D. in Computer Science (GPA: 4.0/4.0)

California, USA

Fall 2025 – Present

Tennessee State University

M.Sc. in Computer Science, Data Science Specialization (GPA: 4.0/4.0)

Tennessee, USA

Fall 2023 – Spring 2025

- **Thesis:** Improving Structured Data Imputation Using Attention Mechanisms.

Brac University

B.Sc. in Computer Science and Engineering (CGPA: 3.87/4.0, Highest Distinction)

Dhaka, Bangladesh

2017 – 2021

- **Thesis:** Facial Expression Recognition: Convolutional Attentional Masking Network and Ensemble Approach.

Research Experience

Graduate Researcher

University of California, Davis

Fall 2025 – Present

- Building a multi-omics framework on the **UK Biobank (~500k participants)** to identify diabetic-retinopathy (DR) resilience, defining DR vs. DR-resistant cohorts (~52k diabetics) from ICD-coded EHR with matched genotype, metabolomic, and clinical data.
- Computed DR and gut-microbiome **polygenic risk scores** (LDpred2-auto over **487,409 imputed genomes**) and integrated 249 NMR plasma-metabolite measures; training penalized-regression and gradient-boosting models for DR-resilience prediction on the UKB Research Analysis Platform (DNAnexus).

Research Associate

CIDA Lab, Tennessee State University

May'2025 – Aug'2025

- Extended end-to-end LLM finetuning for cross-dataset tabular transfer learning, outperforming state-of-the-art ML/DL baselines on 10 benchmark datasets (<10 features) at a fraction of the compute of API-based solutions (*ICECE 2024*).
- Designed transformer-based domain-generalization architectures for robust representation learning across heterogeneous clinical/EHR distributions.

Graduate Research Assistant

CIDA Lab, Tennessee State University

Aug'2023 – May'2025

- Proposed **DeepIFSAC**, a between-feature/between-sample attention imputation framework with CutMix contrastive learning; **outperformed 9 state-of-the-art imputers** across 3 missing-value types (MCAR/MAR/MNAR) and 10–50% missing rates on 12 OpenML datasets and a real EHR dataset, ranking best in 8 of the scenarios at 50% missingness (*Knowledge-Based Systems, 2025*).
- Built the precursor attention-based EHR imputation model (self- vs. inter-sample attention with contrastive data manipulation), achieving the lowest NRMSE on the majority of clinical benchmarks (*IEEE ICHI 2024*).
- Developed a contrastive domain-adaptation method minimizing KL-divergence between source/target distributions for inter- and intra-class alignment (*ICISPC 2024*).

Selected Publications

13 peer-reviewed publications; 75 citations, h-index 6 (**Google Scholar**). **Kowsar, I.** = first author.

- **Kowsar, I.**, Rezaei, S., Meng, F., & Liu, X. (2026). *A Benchmark Study on the Reliability of Explainability Methods*. **CVPR Workshop on Explainable AI for Computer Vision (XAI4CV)**.
- **Kowsar, I.**, Rabbani, S. B., Akhter, K. F. B., et al. (2025). *Deep Cluster Distribution Alignment in Source-Target Domain Adaptation*. **IEEE 4th Int. Conf. on Computing and Machine Intelligence (ICMI)**.
- **Kowsar, I.**, Rabbani, S. B., Hou, Y., & Samad, M. D. (2025). *DeepIFSAC: Deep Imputation of Missing Values Using Feature and Sample Attention within Contrastive Framework*. **Knowledge-Based Systems**, 318, 113506. DOI: 10.1016/j.knosys.2024.113506.

- Samad, M. D., Akhter, K. F. B., Rabbani, S. B., & **Kowsar, I.** (2026). *Imputation-free Learning of Tabular Data with Missing Values using Incremental Feature Partitions in Transformer*. Applied Soft Computing. [link] — outperforms 11 SOTA methods across 17 datasets.
 - **Kowsar, I.**, Rabbani, S. B., & Samad, M. D. (2024). *Attention-based Imputation of Missing Values in Electronic Health Records Tabular Data*. 12th IEEE Int. Conf. on Healthcare Informatics (ICHI). DOI: 10.1109/ICHI61247.2024.00030.
 - **Kowsar, I.**, Rabbani, S. B., Akhter, K. F. B., & Samad, M. D. (2024). *Contrastive Domain Adaptation by Minimizing Divergence in Source-Target Image Distributions*. Int. Conf. on Imaging, Signal Processing and Communications (ICISPC). DOI: 10.1109/ICISPC63824.2024.00018.
 - Rabbani, S. B., **Kowsar, I.**, & Samad, M. D. (2024). *Transfer Learning of Tabular Data by Finetuning Large Language Models*. 13th Int. Conf. on Electrical and Computer Engineering (ICECE).
 - Islam, M. M., Das, A., **Kowsar, I.**, et al. (2021). *Towards Building a Bangla Text Recognition Solution with a Multi-Headed CNN Architecture*. IEEE Int. Conf. on Big Data, pp. 1061-1067. DOI: 10.1109/BigData52589.2021.9671653.
- Preprints / under review:** **Kowsar, I.**, Akhter, K. F., & Samad, M. D. (2025). *LLM Attention Transplant for Transfer Learning of Tabular Data Across Disparate Domains* (arXiv:2511.06161); Akhter, K. F. B., **Kowsar, I.**, & Samad, M. D. *Cross-domain Attention for Transfer Learning between Tabular Data without Shared Features*.

Technical Skills

Languages: Python, Java, SQL

ML / DL Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Hugging Face Transformers

Methods: Attention/Transformers, LLMs, representation & contrastive learning, domain adaptation, imputation

Data & Scientific Computing: Pandas, NumPy, SciPy, Seaborn, Matplotlib, MATLAB, Simulink

Tools & Deployment: Docker, Git, FastAPI, Flask, AWS (EC2, S3), GCP, Jupyter

Databases: MySQL, MongoDB, NoSQL

Teaching & Professional Experience

Graduate Teaching Assistant

Jan'2026 – Jun'2026

University of California, Davis

- Mentored 21 senior students in ECS 193A/B Capstone, guiding end-to-end development of deployable software with academic and industry clients.
- Held weekly office hours advising on system design and implementation; reviewed technical documentation and final presentations.

Machine Learning Engineer

Jul'2021 – Jan'2023

Apurba Technologies Inc.

Sunnyvale, CA

- **Optimized Bengali OCR systems:** designed a multi-headed CNN that reduces multi-class character recognition to a 3-head architecture, reaching state-of-the-art accuracy across handwritten and printed (computer-composed, letterpress, typewritten) text (*IEEE Big Data 2021; ICPR 2022*).
- **Built scalable ML services:** containerized models with Docker and deployed APIs (Flask, FastAPI), reducing inference time by **30%**.

Lecturer

Oct'2021 – Aug'2023

BRAC University

Dhaka, Bangladesh

- Taught Introduction to Robotics (CSE461), System Analysis & Design (CSE471), Digital Logic Design (CSE260), Database Systems (CSE370), and Programming Language I & II.
- Mentored students through the full software development lifecycle, from conceptualization to implementation.

Undergraduate Teaching Assistant

Jan'2020 – May'2021

Programming Language I (Structured) & II (OOP), BRAC University

- Created video tutorials on Object-Oriented Programming in Java & Python; ran consultation hours for problem solving and exam prep.

Certificates & Awards

Highest Distinction, Brac University	2021
Merit Scholarship Award, Brac University	2019–2021
VC's List and Dean's List Award, Brac University	2019–2021
Presentation Skill Award, Brac University	2017